WORKED SOLUTION

STEP I 2001, Question 4

Pure • Inverse Trigonometric Equations

STEP I 2001, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q4

The key idea

The triple-angle formula comes straight from the addition formula for tan. After that, everything hinges on one fact: tan repeats with period π, so if $\tan 3\varphi = \tan 3\theta$ then φ can differ from θ by any multiple of $\pi/3$. The real work in (i) and (ii) is keeping track of the ranges of $\cos^{-1}$ and $\tan^{-1}$, which decide which of those infinitely many angles are allowed.

Solution

The triple-angle formula

Write $t = \tan\theta$. By the addition formula, $\tan 2\theta = \dfrac{2t}{1-t^2}$, and then

$$\tan 3\theta = \tan(2\theta+\theta) = \frac{\tan 2\theta + t}{1 - t\tan 2\theta} = \frac{\dfrac{2t}{1-t^2} + t}{1 - \dfrac{2t^2}{1-t^2}}$$

$$= \frac{2t + t(1-t^2)}{(1-t^2) - 2t^2} = \frac{3t - t^3}{1 - 3t^2},$$

where we multiplied top and bottom by $1 - t^2$. This is the required formula. (It holds whenever both sides are defined.)

The value of $\tan 3\theta$

Since $\cos\theta = 2/\sqrt{5}$ and $0 < \theta < \pi/2$, we have $\sin\theta > 0$, so

$$\sin\theta = \sqrt{1 - \tfrac{4}{5}} = \frac{1}{\sqrt{5}}, \qquad \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1}{2}.$$

(Picture a right-angled triangle with sides 1, 2, $\sqrt{5}$.) Substituting $t = \frac{1}{2}$:

$$\tan 3\theta = \frac{\frac{3}{2} - \frac{1}{8}}{1 - \frac{3}{4}} = \frac{11/8}{1/4} = \frac{11}{2},$$

as required.

It will matter later that θ is fairly small: $\tan\theta = \frac{1}{2} < \frac{1}{\sqrt{3}} = \tan\frac{\pi}{6}$, so $0 < \theta < \pi/6$ and hence $0 < 3\theta < \pi/2$.

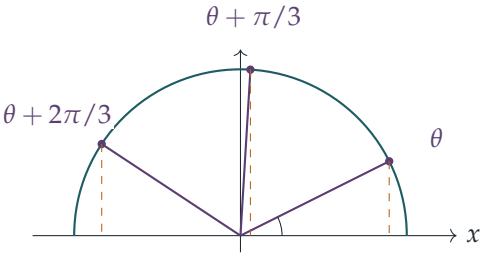
Part (i)

Let $\varphi = \cos^{-1} x$, so $0 \le \varphi \le \pi$ and $x = \cos\varphi$. The equation says $\tan 3\varphi = \frac{11}{2} = \tan 3\theta$. Because tan has period π and takes each value exactly once on each interval of length π,

$$3\varphi = 3\theta + k\pi \quad\Longleftrightarrow\quad \varphi = \theta + \frac{k\pi}{3}, \qquad k \in \mathbb{Z}.$$

Ask yourself How many of these angles are actually allowed?

We need $0 \le \varphi \le \pi$. Since $0 < \theta < \pi/6$, the values $k = 0, 1, 2$ give $\varphi = \theta,\ \theta + \frac{\pi}{3},\ \theta + \frac{2\pi}{3}$, all strictly inside $(0, \pi)$; $k = 3$ gives $\varphi > \pi$ and $k = -1$ gives $\varphi < 0$. So there are exactly three solutions.



The three values of x are the cosines of these angles (the feet of the dashed lines). Using $\cos\theta = 2/\sqrt{5}$, $\sin\theta = 1/\sqrt{5}$ and the addition formula:

$$x_1 = \cos\theta = \frac{2}{\sqrt{5}},$$

$$x_2 = \cos\theta\cos\tfrac{\pi}{3} - \sin\theta\sin\tfrac{\pi}{3} = \frac{2}{\sqrt{5}}\cdot\frac{1}{2} - \frac{1}{\sqrt{5}}\cdot\frac{\sqrt{3}}{2} = \frac{2-\sqrt{3}}{2\sqrt{5}},$$

$$x_3 = \cos\theta\cos\tfrac{2\pi}{3} - \sin\theta\sin\tfrac{2\pi}{3} = -\frac{2}{\sqrt{5}}\cdot\frac{1}{2} - \frac{1}{\sqrt{5}}\cdot\frac{\sqrt{3}}{2} = -\frac{2+\sqrt{3}}{2\sqrt{5}}.$$

None of these angles makes 3φ an odd multiple of $\pi/2$ (each 3φ is $3\theta + k\pi$ with $0 < 3\theta < \pi/2$), so $\tan 3\varphi$ is defined in every case. The solutions of (i) are

$$x = \frac{2}{\sqrt{5}}, \qquad x = \frac{2-\sqrt{3}}{2\sqrt{5}}, \qquad x = -\frac{2+\sqrt{3}}{2\sqrt{5}}.$$

(Rationalising, these are $\frac{2\sqrt{5}}{5}$, $\frac{(2-\sqrt{3})\sqrt{5}}{10}$ and $-\frac{(2+\sqrt{3})\sqrt{5}}{10}$.)

Another route. Putting $T = \tan\varphi$ into the triple-angle formula gives $2(3T - T^3) = 11(1 - 3T^2)$, that is,

$$2T^3 - 33T^2 - 6T + 11 = 0.$$

We know $T = \frac{1}{2}$ is a root, and factorising gives $(2T-1)(T^2 - 16T - 11) = 0$, so $T = \frac{1}{2}$ or $T = 8 \pm 5\sqrt{3}$. Then $x = \cos\varphi = \pm 1/\sqrt{1+T^2}$, with the sign of T (positive T means φ is acute; negative means obtuse). This gives the same three answers, and it is a nice check that $1 + (8+5\sqrt{3})^2 = 140 + 80\sqrt{3} = 1/x_2^2$.

Part (ii)

Let $\psi = \frac{1}{3}\tan^{-1} y$. Since $-\frac{\pi}{2} < \tan^{-1} y < \frac{\pi}{2}$, we have $-\frac{\pi}{6} < \psi < \frac{\pi}{6}$.

Ask yourself Does the condition $\cos\psi = 2/\sqrt{5}$ give one angle or two?

On $(-\pi/6, \pi/6)$, cos is even, so $\cos\psi = 2/\sqrt{5} = \cos\theta$ gives $\psi = \theta$ or $\psi = -\theta$, and both lie in the allowed range because $0 < \theta < \pi/6$. (There are no others: on $[0, \pi/6)$ the function cos is strictly decreasing, so it takes the value $2/\sqrt{5}$ only once there, and similarly on $(-\pi/6, 0]$.)

So $\tan^{-1} y = \pm 3\theta$. Since $0 < 3\theta < \pi/2$, both $\pm 3\theta$ are genuinely values of $\tan^{-1}$, and

$$y = \tan(\pm 3\theta) = \pm\frac{11}{2}.$$

The solutions of (ii) are $y = \frac{11}{2}$ and $y = -\frac{11}{2}$.

Checks

- $\theta = \cos^{-1}(2/\sqrt{5}) \approx 0.4636$, so $3\theta \approx 1.3909$ and $\tan 3\theta \approx 5.5000$.
- $x_2 = \frac{2-\sqrt{3}}{2\sqrt{5}} \approx 0.05992$ and $x_3 \approx -0.83451$; a calculator gives $\tan(3\cos^{-1} x) \approx 5.5$ for each, and a fine numerical scan of $[-1, 1]$ finds exactly three solutions.
- $\cos\left(\frac{1}{3}\tan^{-1}(\pm 5.5)\right) = \cos(\pm 0.4636) \approx 0.8944 = 2/\sqrt{5}$.

Common slips

- Giving only $x = 2/\sqrt{5}$ in (i), forgetting that tan repeats every π, so 3φ has three possible values in $[0, 3\pi]$.
- In (ii), giving only $y = 11/2$: cos is even, so $\psi = -\theta$ works too.
- Not checking ranges: the answers depend on $\theta < \pi/6$, which is why $\theta + 2\pi/3$ is allowed in (i) and why 3θ lies within the range of $\tan^{-1}$ in (ii).

Beyond STEP

Inverse trigonometric functions are a first taste of the problem of inverting a function that isn't one-to-one: we choose a *principal branch*, and equations like these are really about which branch we're on. In complex analysis the same issue appears for $\log z$ and $z^{1/3}$, where going round the origin moves you from one branch to the next, exactly like adding $2\pi/3$ to θ here. Needham's *Visual Complex Analysis* explains branches beautifully with pictures.

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